生物统计和生物信息引论

Lecture 6

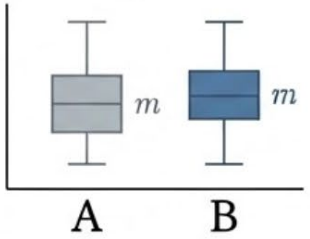
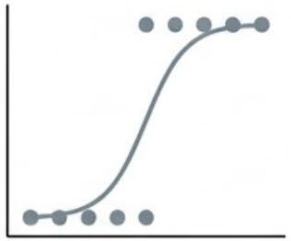
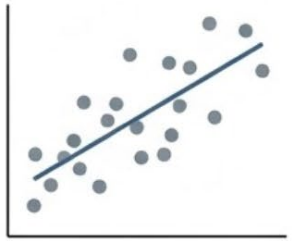
逻辑回归与广义线性模型

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Recall: Predictor vs. Outcome

Your statistical method is strictly dictated by the data types of your Predictor X and Outcome Y.

PREDICTOR (X) \ OUTCOME (Y)	Categorical	Continuous
Categorical	Categorical X, Categorical Y • Contingency Table Analysis 	Categorical X, Continuous Y • ANOVA (F) 
Continuous	Continuous X, Categorical Y • Logistic Regression • Discriminant Analysis 	Continuous X, Continuous Y • Correlation (r) • Linear Regression (R^2) 

Notes: R^2 is the coefficient of determination; F denotes the test statistic.
This matrix provides a general methodological guide based on data types.

二分类结局的建模需求

In many situation, the response variable has only two possible outcomes

- **Disease ($Y = 1$)** vs **Not diseased ($Y = 0$)**
- **Employed ($Y = 1$)** vs **Unemployed ($Y = 0$)**

Response is binary or dichotomous, can model response using Bernoulli distribution:

Y_i	Probability
1	$\Pr(Y_1 = 1) = \pi_i$
0	$\Pr(Y_1 = 0) = 1 - \pi_i$

- $E(Y_i) = \pi_i$
- $Var(Y_i) = \pi_i(1 - \pi_i)$

二分类结局的建模需求

逻辑回归示例数据集：疾病与连续因素关系（虚构数据）

受试者编号	是否患病 (0=无，1=有)	年龄 (岁)	体重指数 (BMI，kg/m^2)	吸烟量 (包/天)	某种生物标志物浓度 (ng/mL)
101	0	28	21.5	0	3.2
102	1	55	29.8	1.5	12.1
103	0	42	24.1	0.5	5.8
104	1	61	31.5	2.0	14.5
105	0	58	21.5	0.5	3.2
106	0	48	29.4	0.5	4.6
107	0	47	27.5	0.5	7.1
108	1	61	23.5	2.0	14.5
109	0	46	29.8	1.0	5.8
100	0	47	27.5	1.0	4.4
101	0	43	23.0	1.0	3.2
102	1	55	29.8	1.5	7.9
103	1	60	29.1	1.0	4.2
104	1	58	29.2	1.0	7.8
105	0	42	31.5	1.0	3.2

🤔 **思考：为什么不能用传统的线性模型去拟合这些数据？它能预测出合理的概率吗？**

二分类结局的建模需求

Goal: express E(Y) as function of a covariate X

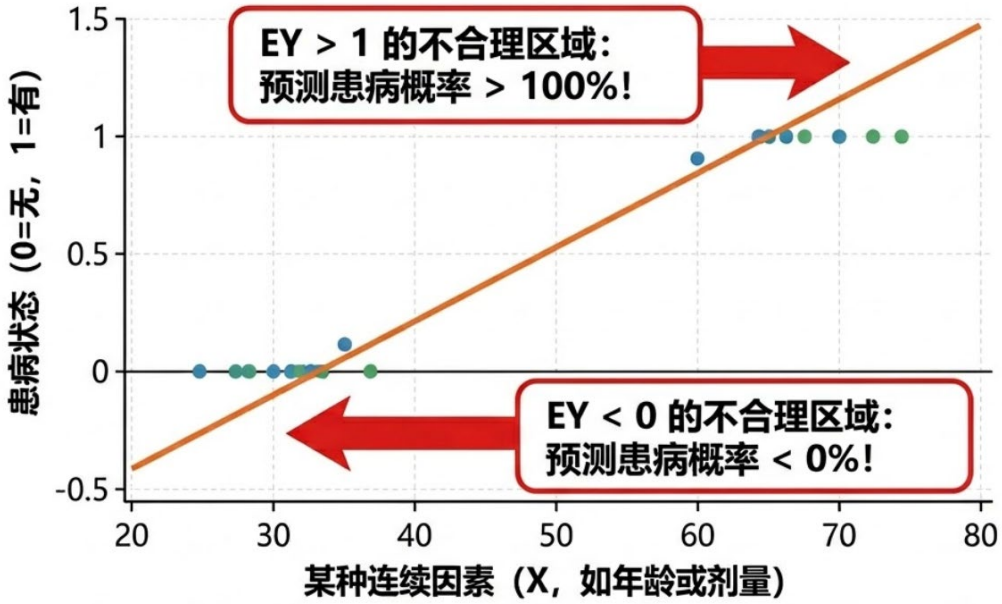
The simple regression is not appropriate

$$E(Y_i) = \beta_0 + \beta_1 X_i$$

It violates several assumptions:

- Does not enforce the constraint $0 \leq E(Y_i) \leq 1$
- Non-normal (binary) distribtion of $\epsilon|X$:

$$\epsilon_i = 0 - \beta_0 - \beta_1 X_i, \text{ when } Y_i = 0$$
$$\epsilon_i = 1 - \beta_0 - \beta_1 X_i, \text{ when } Y_i = 1$$



- Non-constant variance

$$Var(Y_i) = \pi_i(1 - \pi_i) = (\beta_0 + \beta_1 X_i)(1 - \beta_0 - \beta_1 X_i)$$

Logistic 回归模型 - Probability to Odds

- The response variable $Y_i \sim \text{Bernoulli}(\pi_i)$
- **The goal** is to link the expected value $E(Y_i) = \pi_i$ with the predictor X_i
- We need a mathematical **transformation** that maps the interval $[0, 1]$ onto the entire real number line $(-\infty, +\infty)$, allowing us to safely apply a linear fit.
- Recall: In biostatistics and epidemiology, we often use "odds" rather than probability to measure the likelihood of an event.

$$\text{Odds}_i = \frac{\pi_i}{1 - \pi_i}$$

As $\pi_i \to 0$, $\text{Odds}_i \to 0$.

As $\pi_i \to 1$, $\text{Odds}_i \to +\infty$.

We expand the range from $[0, 1]$ to $(0, +\infty)$.
The upper bound is removed, but it still cannot take negative values!

Logistic 回归模型 - Odds to Log-Odds

- The Logit Transformation:

$$\text{Logit}(\pi_i) = \log(\text{Odds}_i) = \log\left(\frac{\pi_i}{1-\pi_i}\right)$$

As $\pi_i \to 0$, $\text{Odds}_i \to 0$, and therefore $\log_e(\text{Odds}_i) \to -\infty$.

As $\pi_i \to 1$, $\text{Odds}_i \to +\infty$, and therefore $\log_e(\text{Odds}_i) \to +\infty$.

- Perfectly maps the original probability to the $(-\infty, +\infty)$ range.

Logistic 回归模型

- Model Construction:

$$\log(\frac{E(Y_i)}{1-E(Y_i)}) = \log(\frac{\pi_i}{1-\pi_i}) = \beta_0 + \beta_1 X_i$$

- Variable Definitions:
  - Y_i : The binary outcome variable (e.g., 1=Disease, 0=Healthy)
  - π_i : The probability of the event occurring given X_i, $P(Y_i = 1|X_i)$.
  - X_i : The predictor variable.
  - β_0, β_1: The model parameters to be estimated.

Logistic 回归模型 - Interpretation of Log-Odds

$$\log(\text{Odds}(X_i)) = \log\left(\frac{\pi_i(X_i)}{1-\pi_i(X_i)}\right) = \beta_0 + \beta_1 X_i$$

- Meaning of β_1 :
  - β_1 is the slope of the true logistic response.
  - It represents the absolute change in the log-odds of the outcome for a one-unit increase in the predictor X_i.

Logistic 回归模型 - Interpretation of Odds Ratio

$$\log(\text{Odds}(X_i)) = \log\left(\frac{\pi_i(X_i)}{1-\pi_i(X_i)}\right) = \beta_0 + \beta_1 X_i$$

- Calculate the difference in log-odds for a one-unit increase in X_i :

$$\log(\text{Odds}(X_i+1)) - \log(\text{Odds}(X_i)) = (\beta_0 + \beta_1(X_i+1)) - (\beta_0 + \beta_1 X_i) = \beta_1$$

$$\text{OR} = \frac{\text{Odds}(X_i+1)}{\text{Odds}(X_i)} = \exp(\beta_1)$$

- **Interpretation**: $\exp(\beta_1)$ represents the true multiplicative change in the odds for a one-unit increase in the predictor, while other predictors are held fixed.

log-odds 与 OR 的解释

- Prospective study: fix predictors, observe the outcomes
  - Example: recruit patients with 2 genotypes, compare occurrence of disease
  - Expensive; large variance for rare diseases

- Retrospective (case/control) study: fix outcome, observe predictors
  - Example: recruit patients with and without the disease,compare the genotypes
  - Cheaper

- Log odds ratio based on logistic regression is the same for both studies.

Logistic 回归模型 - 多元 Logistic

$$\mathbf{X}_i = (1, X_{i1}, X_{i2}, \ldots, X_{i,p-1})^T \qquad \beta = (\beta_0, \beta_1, \beta_2, \ldots, \beta_{p-1})^T$$

- Extension of the logistic response function:

$$E\{Y_i\} = \frac{\exp(\mathbf{X}_i'\beta)}{1 + \exp(\mathbf{X}_i'\beta)}$$

- Extension of the logit link function

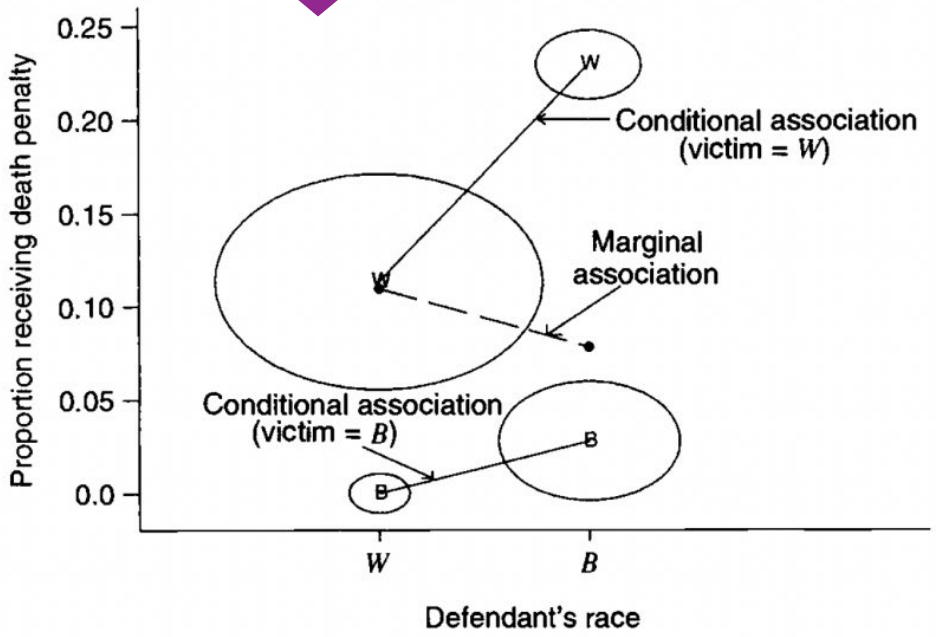
$$\log_e\left(\frac{E\{Y_i\}}{1 - E\{Y_i\}}\right) = \log_e\left(\frac{\pi_i}{1 - \pi_i}\right) = \mathbf{X}_i'\beta$$

- Interpretation of β_j :

$$\log_e\left(\frac{\mathsf{odds}(X_j + 1)}{\mathsf{odds}(X_j)}\right)$$ while other predictors are held fixed

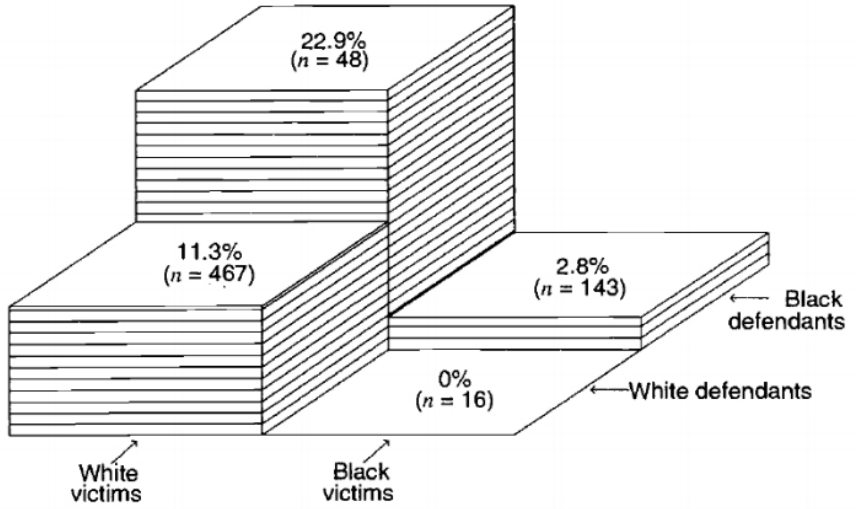
辛普森悖论 (Simpson's Paradox)

Victims' Race	Defendant's Race	Death Penalty: Yes	Death Penalty: No	Percent Yes
White	White	53	414	11.3
	Black	11	37	22.9
Black	White	0	16	0.0
	Black	4	139	2.8
Total	White	53	430	11.0
	Black	15	176	7.9



总体： 白人被告被判死刑的总比例（11.0%）表面上高于黑人被告（7.9%）。

分层： 当引入“受害者种族”进行分层，无论受害者是白人还是黑人，黑人被告被判死刑的比例实际上都更高（22.9% > 11.3%，2.8% > 0%）。



Whites appear more likely to receive the death penalty than Blacks. But is this the whole story?

imbalanced sample distributions!

Parameter interpretation: Univariate and multivariate

Univariate regression: Death Penalty ~ θDefendant Race

Multivariate regression: Death Penalty ~ β_1Defendant Race + β_2 Victim Race

Are β_1 and θ have the same sign? (effects direction are the same)?

Fitted probability

- Logisc regression also helps us to predict the probability π_i.
- We can solve for it by reversing the equation

$$\frac{\pi_i}{1-\pi_i} = \exp(\beta_0 + \beta_1 X_i)$$

$$E(Y_i) = \frac{\exp(\beta_0 + \beta_1 X_i)}{1 + \exp(\beta_0 + \beta_1 X_i)} = \frac{1}{1 + \exp(-\beta_0 - \beta_1 X_i)}$$

This is the sigmoidal response function

- A monotonic increasing/decreasing function
- Restricts $0 \leq E(Y_i) \leq 1$
- Symmetry property
- Example of a *nonlinear* model

Logistic 回归模型 - 概率分布

Y_i are *independent but not identically* distributed Bernoulli random variables

$$Y_i \overset{ind}{\sim} Bernoulli(\pi_i) \ where$$

$$\pi_i = \frac{\exp(\beta_0 + \beta_1 X_i)}{1 + \exp(\beta_0 + \beta_1 X_i)}$$

Note *no more error term*!

Probability density of Y_i

$$f(Y_i) = \pi_i^{Y_i}(1 - \pi_i)^{1-Y_i}$$

Least squares estimates are inappropriate, use maximum likelihood for parameter estimation

Logistic 回归模型 - 极大似然估计

$Y_i \overset{ind}{\sim} \text{Bernoulli}(\pi_i)$, where $\pi_i = \frac{\exp(\beta_0+\beta_1 X_i)}{1+\exp(\beta_0+\beta_1 X_i)}$

- Log likelihood

$$\begin{aligned}
\log_e(L) &= \log\{\prod_{i=1}^{n} \pi_i^{Y_i}(1-\pi_i)^{1-Y_i}\} \\
&= \sum_{i=1}^{n} Y_i \log(\pi_i) + \sum_{i=1}^{n}(1-Y_i)\log(1-\pi_i) \\
&= \sum_{i=1}^{n} Y_i \log(\frac{\pi_i}{1-\pi_i}) + \sum_{i=1}^{n}\log(1-\pi_i) \\
&= \sum_{i=1}^{n} Y_i(\beta_0+\beta_1 X_i) - \sum_{i=1}^{n}\log(1+\exp(\beta_0+\beta_1 X_i))
\end{aligned}$$

$$b = (b_0, b_1) = \arg\max_{\beta_0,\beta_1} \log_e(L)$$

$$\hat{\pi}_i = \frac{\exp(b_0+b_1 X_i)}{1+\exp(b_0+b_1 X_i)}$$

- MLEs do not have closed forms: Newton/ Fisher scoring/ Gradient Descent

Newton's Method — Finding the MLE Iteratively

Logistic regression log-likelihood is **concave but has no closed-form solution** for $\hat{\beta}$ —we must optimize numerically.

Key idea: approximate the score equation $\ell'(\beta) = 0$ with a **linear (Taylor) expansion** at the current iterate $\beta^{(t)}$, then solve the linearized equation exactly.

Updating rule (univariate intuition): for target function $g(x)$

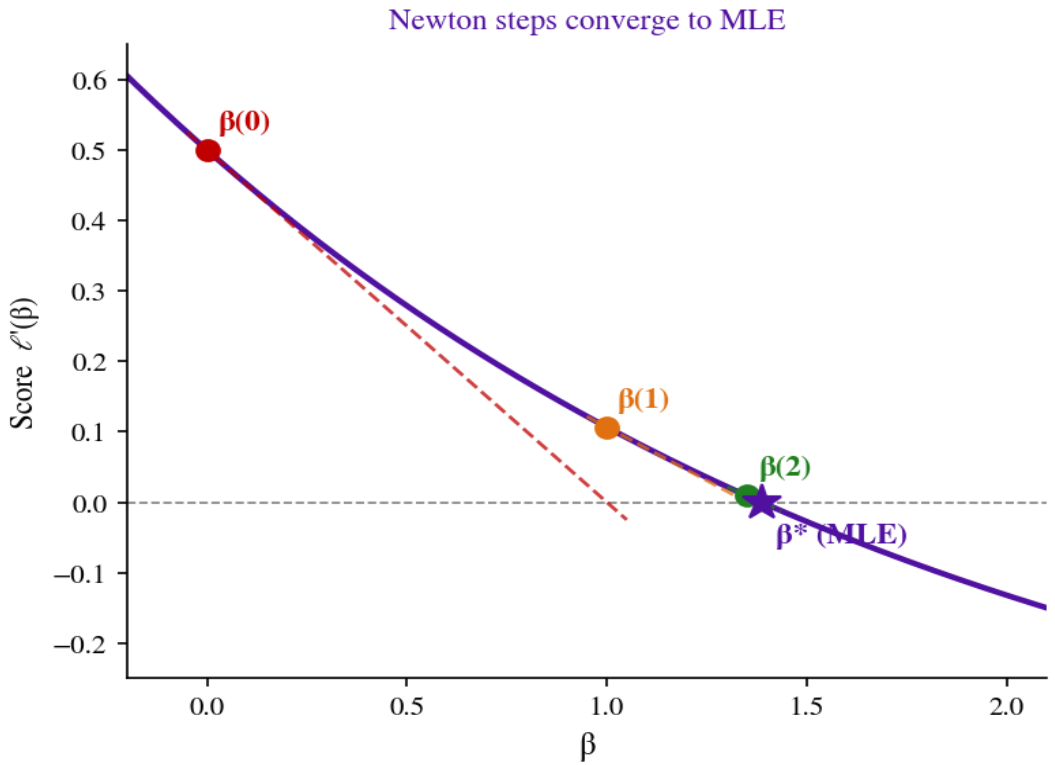
$$x^{(t+1)} = x^{(t)} - \frac{g'(x^{(t)})}{g''(x^{(t)})}$$

Multivariate extension ($\beta \in \mathbb{R}^p$):

$$\boldsymbol{\beta}^{(t+1)} = \boldsymbol{\beta}^{(t)} - [\ell''(\boldsymbol{\beta}^{(t)})]^{-1}\ell'(\boldsymbol{\beta}^{(t)})$$

$\ell'(\beta)$ = score (gradient of log-likelihood)

$\ell''(\beta)$ = Hessian (matrix of second derivatives)



https://en.wikipedia.org/wiki/Newton%27s_method

Fisher Scoring — From Newton's Method to IRLS

- **Fisher Scoring**: replace the observed Hessian $\ell''(\beta)$ in Newton's method with its **expectation — the Fisher Information matrix I(β).**

$$\boldsymbol{\beta}^{(t+1)} = \boldsymbol{\beta}^{(t)} + \mathbf{I}(\boldsymbol{\beta}^{(t)})^{-1}\, \ell'(\boldsymbol{\beta}^{(t)})$$

For **logistic regression**:
Hessian does **not depend on Y** → observed information = Fisher information → Newton = Fisher Scoring.

- **Iteratively Reweighted Least Squares:** solve a *weighted* least squares problem with updated weights $W^{(t)}$ and working response $z^{(t)}$, then re-compute weights from the new $\hat{\beta}$ and repeat until convergence.

Applying Fisher Scoring to the logistic log-likelihood yields **IRLS** at each iteration:

$$\boldsymbol{\beta}^{(t+1)} = (X^\top W^{(t)} X)^{-1} X^\top W^{(t)} \mathbf{z}^{(t)}$$

$W^{(t)} = diag\{\pi_i(1-\pi_i)\}$ (logistic variance, from Fisher information)
$z^{(t)} = X\beta^{(t)} + W^{-1}(y-\pi^{(t)})$ (working response)

Take-away: **Newton's Method** finds roots of score equations iteratively. **Fisher Scoring** stabilises it by using expected curvature. For logistic regression these two are identical, and both reduce to **IRLS** — which is what R's **glm()** runs under the hood.

Newton's Method and Fisher Scoring


```r
# 1. 构建模拟二分类数据集
set.seed(42)
n <- 500
X <- rnorm(n, mean = 0, sd = 1)
true_b0 <- -0.5
true_b1 <- 1.5
p <- 1 / (1 + exp(-(true_b0 + true_b1 * X)))
Y <- rbinom(n, size = 1, prob = p)
my_data <- data.frame(X = X, Y = Y)

# ============================================================================
# 方法一：使用 glm 函数（底层算法：IRLS / Fisher Scoring 二阶优化）
# ============================================================================
cat("\n--- 方法一：glm() 函数 (Fisher Scoring) ---\n")
model_glm <- glm(Y ~ X, data = my_data, family = binomial(link = "logit"))

cat("1. 估计的参数 [b0, b1]:\n")
print(coef(model_glm))

cat("\n2. 收敛所需迭代次数 (Fisher Scoring iterations):\n")
print(model_glm$iter)
```

```
--- 方法一：glm() 函数 (Fisher Scoring) ---
> model_glm <- glm(Y ~ X, data = my_data, family = binomial(link = "logit"))
>
> cat("1. 估计的参数 [b0, b1]:\n")
1. 估计的参数 [b0, b1]:
> print(coef(model_glm))
(Intercept)           X
 -0.2477624   1.6521122
>
> cat("\n2. 收敛所需迭代次数 (Fisher Scoring iterations):\n")

2. 收敛所需迭代次数 (Fisher Scoring iterations):
> print(model_glm$iter)
[1] 5
```

单步计算量高、迭代步数少，不适合变量多的数据
强调参数的可解释性和统计显著性

Gradient Descent

为避免计算和求解高维海森矩阵（Hessian）的巨大计算开销，直接用负单位矩阵及标量步长替代复杂的二阶导数信息

$$x^{(t+1)} = x^{(t)} + \alpha^{(t)} g'(x^{(t)})$$

$\alpha^{(t)} > 0$ 为**步长参数**，用于控制每次顺着梯度方向前移的步幅

```r
# ============================================================================
# 方法二：使用 optim 函数（底层算法：BFGS 拟牛顿法/基于梯度的优化）
# ============================================================================
cat("\n\n--- 方法二：optim() 函数（BFGS 算法）---\n")

# 定义目标函数：负对数似然
neg_log_likelihood <- function(beta) {
  b0 <- beta[1]
  b1 <- beta[2]
  p_hat <- 1 / (1 + exp(-(b0 + b1 * my_data$X)))
  # 数值截断，防止 log(0)
  p_hat <- pmax(pmin(p_hat, 1 - 1e-15), 1e-15)
  nll <- -sum(my_data$Y * log(p_hat) + (1 - my_data$Y) * log(1 - p_hat))
  return(nll)
}

optim_result <- optim(par = c(0, 0),
                      fn = neg_log_likelihood,
                      method = "BFGS")

cat("1. 估计的参数 [b0, b1]:\n")
params_optim <- optim_result$par
names(params_optim) <- c("(Intercept)", "X")
print(params_optim)

cat("\n2. 收敛所需计算次数 (Function and Gradient evaluations):\n")
print(optim_result$counts)
```

```
> cat("1. 估计的参数 [b0, b1]:\n")
1. 估计的参数 [b0, b1]:
> params_optim <- optim_result$par
> names(params_optim) <- c("(Intercept)", "X")
> print(params_optim)
(Intercept)           X
 -0.2477624   1.6521123
>
> cat("\n2. 收敛所需计算次数 (Function and Gradient evaluations):\n")

2. 收敛所需计算次数 (Function and Gradient evaluations):
> print(optim_result$counts)
function gradient
      28       10
```

单步计算量低、迭代步数多，适合海量数据或非常多的变量，强调模型的预测精度

Output

- Fitted regression coefficients
- Fitted value for the individual i

$$\hat{\pi}_i = \frac{e^{b_0+b_1X_i}}{1+e^{b_0+b_1X_i}}$$

- Fitted logistic response (i.e. fitted log odds)

$$\log_e(\widehat{odds}(X_i)) = \log_e \frac{\hat{\pi}_i(X_i)}{1-\hat{\pi}_i(X_i)} = b_0 + b_1X_i$$

b_1 is the slope of the fitted logistic response

- b_1 is interpreted as log(odds ratio)

$$b_1 = \log_e\left(\frac{\hat{\pi}_i(X_i+1)}{1-\hat{\pi}_i(X_i+1)}\right) - \log_e\left(\frac{\hat{\pi}_i(X_i)}{1-\hat{\pi}_i(X_i)}\right) = \log_e\left(\frac{\mathsf{odds}(X_i+1)}{\mathsf{odds}(X_i)}\right)$$

$$\widehat{\mathsf{OR}}(X_i) = \exp(b_1)$$

log-odds and OR in logistic regression

- With retrospective sampling and selected indicator $Z_i = 1($
- we model:

Denote the selection probability

$$P(Z_i = 1 \mid Y_i = y, X_i) = \theta_y.$$

Then

$$\begin{aligned} P(Y_i = y, Z_i = 1 \mid X_i) &= P(Z_i = 1 \mid Y_i = y, X_i)\, P(Y_i = y \mid X_i) \\ &= \theta_y\, P(Y_i = y \mid X_i). \end{aligned}$$

Hence,

$$\begin{aligned} P(Y_i = 1, Z_i = 1 \mid X_i) &= \theta_1 \pi(X_i), \\ P(Y_i = 0, Z_i = 1 \mid X_i) &= \theta_0 \big(1 - \pi(X_i)\big). \end{aligned}$$

$$\begin{aligned} & P(Y_i = 1 | X_i, Z_i = 1) \\ &= \frac{P(Y_i = 1, Z_i = 1 | X_i)}{P(Z_i = 1 | X_i)} \\ &= \frac{P(Y_i = 1, Z_i = 1 | X_i)}{P(Y_i = 0, Z_i = 1 | X_i) + P(Y_i = 1, Z_i = 1 | X_i)} \\ &= \frac{\theta_1 \times \pi(X_i)}{\theta_0 \times (1 - \pi(X_i)) + \theta_1 \times \pi(X_i)} \\ &= \frac{\theta_1 e^{\beta_0 + \beta_1 X_i}}{\theta_0 + \theta_1 e^{\beta_0 + \beta_1 X_i}} \end{aligned}$$

- uncover the same β_1 as in the prospective study

$$\Longrightarrow \log\{\frac{P(Y_i = 1 | X_i, Z_i = 1)}{P(Y_i = 0 | X_i, Z_i = 1)}|\} = [\log(\theta_1/\theta_0)] + \beta_1 X_i$$

Logistic 回归模型 - 多元 Logistic

$$\mathbf{X}_i = (1, X_{i1}, X_{i2}, \ldots, X_{i,p-1})^T \qquad \beta = (\beta_0, \beta_1, \beta_2, \ldots, \beta_{p-1})^T$$

- Extension of the logistic response function:

$$E\{Y_i\} = \frac{\exp(\mathbf{X}_i'\beta)}{1 + \exp(\mathbf{X}_i'\beta)}$$

- Extension of the logit link function

$$\log_e\left(\frac{E\{Y_i\}}{1 - E\{Y_i\}}\right) = \log_e\left(\frac{\pi_i}{1 - \pi_i}\right) = \mathbf{X}_i'\beta$$

- The multivariate likelihood function:

$$\log_e L(\beta) = \sum_{i=1}^{n} Y_i(\mathbf{X}_i'\beta) - \sum_{i=1}^{n} \log_e[1 + \exp(\mathbf{X}_i'\beta)]$$

- Interpretation of b_j :

$\log_e\left(\frac{\text{odds}(X_j + 1)}{\text{odds}(X_j)}\right)$ while other predictors are held fixed

Multivariate Logistic Regression

Dataset: *Pima.tr* (MASS package, n = 200)

Y: type — diabetes diagnosis (Yes / No)

X: 6 clinical predictors:

Model:

$$\text{logit}(P(\text{diabetes} = \text{Yes})) = \beta_0 + \beta_1 \cdot npreg + \ldots + \beta_6 \cdot age$$

```r
library(MASS)
data(Pima.tr)

# Fit multivariate logistic regression
fit <- glm(type ~ npreg + glu + bp +
             bmi + ped + age,
           data   = Pima.tr,
           family = binomial(link = "logit"))

# Coefficient summary
summary(fit)

# Odds Ratios + 95% CI
cbind(
  OR  = round(exp(coef(fit)), 3),
  CI  = round(exp(confint(fit)), 3)
)

# p-values
coef_tab <- coef(summary(fit))
round(coef_tab[, "Pr(>|z|)"], 4)
```

```
> summary(fit)

Call:
glm(formula = type ~ npreg + glu + bp + bmi + ped + age, family = binomial(link = "logit"),
    data = Pima.tr)

Coefficients:
             Estimate Std. Error z value Pr(>|z|)
(Intercept) -9.734824   1.711096  -5.689 1.28e-08 ***
npreg        0.103254   0.064693   1.596  0.11048
glu          0.032113   0.006786   4.732 2.22e-06 ***
bp          -0.004961   0.018414  -0.269  0.78763
bmi          0.081327   0.033241   2.447  0.01442 *
ped          1.819249   0.665184   2.735  0.00624 **
age          0.041036   0.022034   1.862  0.06255 .
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 256.41  on 199  degrees of freedom
Residual deviance: 178.40  on 193  degrees of freedom
AIC: 192.4

Number of Fisher Scoring iterations: 5
```

Wald Test

```
                OR 2.5 % 97.5 %
(Intercept) 0.000 0.000  0.001
npreg       1.109 0.978  1.263
glu         1.033 1.020  1.047
bp          0.995 0.959  1.032
bmi         1.085 1.018  1.160
ped         6.167 1.727 23.634
age         1.042 0.998  1.089
```

glu: OR = 1.033

each 1-unit ↑ in glucose → 3.3% ↑ in odds of diabetes

```
> round(coef_tab[, "Pr(>|z|)"], 4)
(Intercept)       npreg         glu          bp         bmi         ped         age
     0.0000      0.1105      0.0000      0.7876      0.0144      0.0062      0.0625
```

Interpretation & Prediction


```r
# Predict for new patients
new_patients <- data.frame(
  npreg = c(1,   5  ),
  glu   = c(90,  160),
  bp    = c(70,  85 ),
  bmi   = c(24,  38 ),
  ped   = c(0.20, 0.90),
  age   = c(25,  50 )
)

# Predicted probabilities
pred <- predict(fit,
                newdata = new_patients,
                type    = "response")
round(pred, 3)

# All training observations
Pima.tr$prob_hat <-
  predict(fit, type = "response")
head(Pima.tr[, c("type","prob_hat")])
```

```
> round(pred, 3)
    1     2
0.023 0.907
>
> # All training observations
> Pima.tr$prob_hat <-
+   predict(fit, type = "response")
> head(Pima.tr[, c("type","prob_hat")])
  type   prob_hat
1   No 0.06350462
2  Yes 0.81720882
3   No 0.07439343
4   No 0.65032744
5   No 0.03684256
6  Yes 0.31781282
```

$$\hat{\eta} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2 + \cdots + \hat{\beta}_p x_p = \mathbf{x}_{new}^T \hat{\boldsymbol{\beta}}$$

$$\hat{P}(Y = 1 | \mathbf{x}_{new}) = \frac{\exp(\hat{\eta})}{1 + \exp(\hat{\eta})} = \frac{1}{1 + \exp(-\hat{\eta})}$$

如果在 glm 拟合的模型中使用默认的 predict()，R 输出的将是线性预测 $\eta = \mathbf{x}^T \boldsymbol{\beta}$

Logistic 回归模型 - b 的渐近性质

- **Asymptotic existence and uniqueness**: In small samples, the MLE might not exist or might not be unique. However, as the sample size n grows, the probability that a unique maximum likelihood estimate b exists approaches 1:

$$P\{b \text{ exists and is unique}\} \to 1 \text{ as } n \to \infty$$

- **Consistency**: The estimator b converges in probability to the true population parameter β. This guarantees that as we collect more data, our estimate will eventually pinpoint the true parameter:

$$b \xrightarrow{p} \beta \text{ as } n \to \infty$$

Logistic 回归模型 - b 的渐近性质

- **Asymptotic Normality**: As n increases, the distribution of the estimator b approaches a multivariate normal distribution centered at the true parameter β:

$$\sqrt{n}(b-\beta) \xrightarrow{d} N(0, \mathbf{I}(\beta)^{-1}) \text{ as } n \to \infty$$

Here, $\xrightarrow{d}$ denotes convergence in distribution, and $\mathbf{I}(\beta)$ represents the **Fisher Information Matrix.**

- **Asymptotic efficiency**: the MLE has asymptotically smaller variance than many other estimators. It achieves the theoretical lower bound of variance (the Cramér-Rao bound) for consistent estimators as the sample size approaches infinity.

Logistic 回归模型 - b 的渐近性质

- To utilize the asymptotic normality for inference, we must quantify the variance $\mathbf{I}(\beta)^{-1}$.
- **Observed Information Matrix:** Since the true β is unknown, we evaluate the negative Hessian matrix (the matrix of second-order partial derivatives) at the estimated value b:

$$\mathbf{I}(b) = \left[-\frac{\partial^2 \log_e L(\beta)}{\partial \beta \partial \beta^T} \right]_{\beta=b}$$

- **Estimated Covariance Matrix**: By inverting this observed information matrix, we obtain the estimated variance-covariance matrix

$$s^2\{b\} = \left(\left[-\frac{\partial^2 \log_e L(\beta)}{\partial \beta_j \partial \beta_j'} \right]_{\beta=b} \right)^{-1}$$

- The estimated variance for a specific coefficient b_j is simply the corresponding j-th element on the main diagonal of this inverted matrix.
- Taking its square root provides the **standard error** $s\{b_j\}$, which directly forms the denominator for the upcoming Wald test statistic.

Recall: Wald Test

Measures how far the estimated parameter MLE $\hat{\theta}$ is from the hypothesized value θ_0, relative to its standard error.

- **Test Statistic(1D)**: $Z_W = \frac{\hat{\theta}-\theta_0}{\sqrt{I^{-1}(\hat{\theta})}} \sim N(0,1) \Rightarrow W_p = Z_W^2 = (\hat{\theta}-\theta_0)^2 I(\hat{\theta}) \sim \chi_1^2]$.
- **Estimation**: Requires calculating the unconstrained MLE $\hat{\theta}$ and estimationg its variance $I(\hat{\theta})$ or observed information $J(\hat{\theta})$.
- **Properties**: Computationally easy (only fit full model). However, it is not invariant to reparameterization and can be inaccurate for small n.

Recall: Likelihood Ratio Test (LRT)

Compares the maximum log-likelihood under the full model $l(\hat{\theta})$ to the maximum log-likelihood under the restricted null model $l(\theta_0)$.

- **Test Statistic**: $W = 2[l(\hat{\theta}) - l(\theta_0)] \sim \chi_1^2]$.
- **Estimation**: Requires fitting the model twice (both under H_1 to get $\hat{\theta}$ and under H_0).
- **Properties**: Highly reliable in small samples. Invariant to reparameterization (e.g., testing θ vs $\log\theta$ give the same result).

Logistic 回归模型 - Wald test

- Test $H_0 : \beta_j = 0$ versus $H_a : \beta_j \neq 0$
- Test statistic $z^* = \frac{b_j - 0}{s\{b_j\}}$
- *Approximate* variance $s^2\{b\}$

$$s^2\{b\} = ([-\frac{\partial^2 \log_e L(\beta)}{\partial \beta_j \partial \beta_j'}]_{\beta=b})^{-1}$$

- *Approximate* distribution of z

$$z^* \sim N(0,1), \text{ alternatively,} (z^*)^2 \sim \chi_1^2$$
Reject H_0 if $|z^*| > z_{1-\alpha/2}$
CI for β_j: $b_j \pm z_{1-\alpha/2} s\{b_j\}$

Logistic 回归模型 - Wald test


```r
# 1. 模拟二分类数据集（临床背景：年龄与是否患病）
set.seed(123)
n <- 200
age <- rnorm(n, mean = 50, sd = 10)
# 设定真实的参数
true_b0 <- -5
true_b1 <- 0.1
# 生成真实概率并抽样
p <- 1 / (1 + exp(-(true_b0 + true_b1 * age)))
disease <- rbinom(n, size = 1, prob = p)
data <- data.frame(age = age, disease = disease)

# 2. 拟合 Logistic 回归模型
model <- glm(disease ~ age, data = data, family = binomial(link = "logit"))

# ========================================
# 直接查看 summary() 中的 Wald 检验结果
# ========================================
cat("\n--- 1. summary() 默认的 Wald 检验输出 ---\n")
model_summary <- summary(model)
print(model_summary$coefficients)
```

```
--- 1. summary() 默认的 Wald 检验输出 ---
> model_summary <- summary(model)
> print(model_summary$coefficients)
               Estimate Std. Error   z value     Pr(>|z|)
(Intercept) -4.03445182  0.9055569 -4.455216 8.380866e-06
age          0.07920233  0.0179097  4.422315 9.764887e-06
```

强烈拒绝原假设 $H_0 : \beta_{\text{age}} = 0$。

这表明年龄（age）对患病概率有显著的正向影响。

Logistic 回归模型 - Likelihood Ratio test

- Multivariate logistic regression

$$\log\left(\frac{E\{Y_i\}}{1-E\{Y_i\}}\right) = \beta_0 + \beta_1 X_{i1} + \cdots + \beta_{p-1} X_{i,p-1}$$

- Test $H_0: \beta_1 = \beta_2 = \cdots = \beta_q = 0$ versus H_a: not all $\beta_1, \beta_2, \cdots, \beta_q = 0$

$$\log L(\text{full}) = \sum_{i=1}^{n} \left[Y_i \log(\hat{\pi}_{i(\text{full})}) + (1-Y_i)\log(1-\hat{\pi}_{i(\text{full})}) \right]$$

$$\log L(\text{reduced}) = \sum_{i=1}^{n} \left[Y_i \log(\hat{\pi}_{i(\text{reduced})}) + (1-Y_i)\log(1-\hat{\pi}_{i(\text{reduced})}) \right]$$

- Test statistic

$$G^2 = 2\sum_{i=1}^{n} \left[Y_i \log\left(\frac{\hat{\pi}_{i(\text{full})}}{\hat{\pi}_{i(\text{reduced})}}\right) + (1-Y_i)\log\left(\frac{1-\hat{\pi}_{i(\text{full})}}{1-\hat{\pi}_{i(\text{reduced})}}\right) \right]$$

- *Approximate* distribution of G^2 for large n, reject H_0 if $G^2 > \chi^2(1-\alpha, q)$

Logistic 回归模型 - Likelihood Ratio test


```r
# ==========================================
# 准备阶段：拟合完整模型与简化模型
# ==========================================
# 完整模型 (Full model)：包含 age 变量 (直接沿用上一个例子的 model)
model_full <- glm(disease ~ age, data = data, family = binomial(link = "logit"))

# 简化模型 (Reduced model)：仅包含截距项 (Null model)
# 对应于检验 H0: beta_1 = 0
model_null <- glm(disease ~ 1, data = data, family = binomial(link = "logit"))

# ==========================================
# 方式一：直接使用 anova() 函数进行 LRT
# ==========================================
cat("\n--- 1. anova() 默认的似然比检验输出 ---\n")
# test = "Chisq" 指示 R 计算卡方检验的 P 值
lrt_result <- anova(model_null, model_full, test = "Chisq")
print(lrt_result)
```

```
--- 1. anova() 默认的似然比检验输出 ---
> # test = "Chisq" 指示 R 计算卡方检验的 P 值
> lrt_result <- anova(model_null, model_full, test = "Chisq")
> print(lrt_result)
Analysis of Deviance Table

Model 1: disease ~ 1
Model 2: disease ~ age
  Resid. Df Resid. Dev Df Deviance Pr(>Chi)
1       199     276.94
2       198     253.76  1   23.174 1.48e-06 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

Logistic 回归模型 - Likelihood Ratio test


```r
# ========================================
# 方式二：根据理论公式手动计算 G^2 统计量
# ========================================
cat("\n--- 2. 手动推导计算 Likelihood Ratio 统计量 ---\n")

# 提取两个模型的对数似然值 (Log-Likelihood)
ll_null <- as.numeric(logLik(model_null))
ll_full <- as.numeric(logLik(model_full))

# 计算似然比检验统计量 G^2
# 公式: G^2 = -2 * [log L(reduced) - log L(full)]
G2_stat <- -2 * (ll_null - ll_full)

# 计算自由度的差值 (参数个数的差值)
df_diff <- df.residual(model_null) - df.residual(model_full)

# 计算 P 值
# G^2 统计量在大样本下渐近服从卡方分布
p_value_lrt <- pchisq(G2_stat, df = df_diff, lower.tail = FALSE)

# 整理手动计算的结果
manual_lrt_results <- data.frame(
  LogLik_Null = ll_null,
  LogLik_Full = ll_full,
  G2_Statistic = G2_stat,
  DF = df_diff,
  P_value = p_value_lrt
)
rownames(manual_lrt_results) <- "LRT (手动计算)"
print(manual_lrt_results)
```

```
> print(manual_lrt_results)
                LogLik_Null LogLik_Full G2_Statistic DF      P_value
LRT (手动计算)    -138.4694   -126.8822     23.17443  1 1.479516e-06
```

Wald Test Versus Likelihood Ratio Test

- Both tests are approximate, for large n.
- Likelihood Ratio test: $\beta_j = 0$ or several $\beta_j = 0$ simultaneously. Note no one-sided test.
- Wald test:$H_0: \beta_j = c$ (where c is a specific constant of interest), or linear combinations of β.
- When testing a single $H_0: \beta_j = 0$, the tests may lead to different conclusions, due to the approximate nature of the tests (unlike linear regression).

Model specification

- model specified by researchers
- Is the specified model correct?
- What method helps us check it? Experiments?

重复数据的拟合不足 – Pearson χ^2

$$H_0 : \log(\frac{E\{Y_{ij}\}}{1 - E\{Y_{ij}\}}) = \beta_0 + \beta_1 X_{i1} + \cdots + \beta_{p-1} X_{i,p-1}$$

$$H_a : \log(\frac{E\{Y_{ij}\}}{1 - E\{Y_{ij}\}}) \neq \beta_0 + \beta_1 X_{i1} + \cdots + \beta_{p-1} X_{i,p-1}$$

- c covariate configurations, each with n_i cases
- Observed counts, O_{i0} and O_{i1} (observed # of 0's and # of 1's in configuration i)
- Expected counts **under the same mean assumption**

$$E_{i0} = n_i(1 - \hat{\pi}_i), E_{i1} = n_i\hat{\pi}_i$$

- Test statistic: reject H_0 if

$$\chi^2 = \sum_{i=1}^{c}\sum_{k=0}^{1} \frac{(O_{ik} - E_{ik})^2}{E_{ik}} > \chi^2(1 - \alpha, c - p)$$

重复数据的拟合不足 – 偏差 (Deviance)

$$H_0 : \log(\frac{E\{Y_{ij}\}}{1-E\{Y_{ij}\}}) = \beta_0 + \beta_1 X_{i1} + \cdots + \beta_{p-1} X_{i,p-1}$$

$$H_a : \log(\frac{E\{Y_{ij}\}}{1-E\{Y_{ij}\}}) \neq \beta_0 + \beta_1 X_{i1} + \cdots + \beta_{p-1} X_{i,p-1}$$

- c covariate configurations, each with n_i cases

$$H_0 : \text{model of interest}, \ H_a : \text{saturated model}$$

- Model of interest: $E\{Y_{ij}\} = \pi_i, \widehat{E\{Y_{ij}\}} = \hat{\pi}$
- Saturated model: $E\{Y_{ij}\} = p_i, \widehat{E\{Y_{ij}\}} = \hat{p}_i = \frac{\sum_j Y_{ij}}{n_i}$
- Test statistic: LLR, also called deviance
- Deviance of a saturated model always = 0

重复数据的拟合不足 – 偏差 (Deviance)

$$
\begin{aligned}
G^2 &= DEV(X_0, X_1, \cdots, X_{p-1}) \\
&= -2[\log_e L(\text{current model}) - \log_e L(\text{saturated model})] \\
&= -2\sum_{i=1}^{c}[\sum_{j=1}^{n_i} Y_{ij} \log_e \hat{\pi}_i + (n_i - \sum_{j=1}^{n_i} Y_{ij}) \log_e(1-\hat{\pi}_i)] \\
&+ 2\sum_{i=1}^{c}[\sum_{j=1}^{n_i} Y_{ij} \log_e \hat{p}_i + (n_i - \sum_{j=1}^{n_i} Y_{ij}) \log_e(1-\hat{p}_i)] \\
&= -2\sum_{i=1}^{c}[\sum_{j=1}^{n_i} Y_{ij} \log_e(\frac{\hat{\pi}_i}{\hat{p}_i}) + (n_i - \sum_{j=1}^{n_i} Y_{ij}) \log_e(\frac{1-\hat{\pi}_i}{1-\hat{p}_i})]
\end{aligned}
$$

- Reject H_0 if $G_2 > \chi^2(1-\alpha, c-p)$
- Approximation $\chi^2(1-\alpha, c-p)$ can be poor when n_i are small
- The closer the link to identity, the better the approximation

未重复数据的评估：Hosmer-Lemeshow

$$H_0 : \log(\frac{E\{Y_{ij}\}}{1-E\{Y_{ij}\}}) = \beta_0 + \beta_1 X_{i1} + \cdots + \beta_{p-1} X_{i,p-1}$$

$$H_a : \log(\frac{E\{Y_{ij}\}}{1-E\{Y_{ij}\}}) \neq \beta_0 + \beta_1 X_{i1} + \cdots + \beta_{p-1} X_{i,p-1}$$

- For a few predictors, we can group cases based on Xs.
- Group cases based on values of estimated probabilities $\hat{\pi}_i$ into c groups (E.g. find $c = 9$ groups based on percentiles)
- Apply Pearson χ^2 test to the groups
- Reject H_0 if $X^2 > \chi^2(1-\alpha, c-2)$. It was showed by simulation that this distribution is appropriate

诊断：残差分析

Logistic Regression **Residuals**

$$e_i = \begin{cases} 1 - \widehat{\pi}_i, \text{ if } Y_i = 1 \\ -\widehat{\pi}_i, \text{ if } Y_i = 0 \end{cases}$$

Pearson residual

$$r_{P_i} = \frac{Y_i - \widehat{\pi}_i}{\sqrt{\widehat{\pi}_i (1 - \widehat{\pi}_i)}}$$

- e_i divided by the standard error of Y_i
- $\Sigma_{i=1}^{n} r_{P_i}^2$ equals non-replicated Pearson χ^2

诊断：残差分析

Logistic Regression **Residuals**

$$e_i = \begin{cases} 1 - \widehat{\pi}_i, \text{ if } Y_i = 1 \\ -\widehat{\pi}_i, \text{ if } Y_i = 0 \end{cases}$$

Studentized Pearson Residual

$$r_{SP_i} = \frac{Y_i - \widehat{\pi}_i}{\sqrt{\widehat{\pi}_i \left(1 - \widehat{\pi}_i\right) \cdot \left(1 - h_{ii}\right)}}$$

- e_i divided by the standard error of e_i $\longrightarrow$ unit variance
- h_{ii} is the diagonal element of the hat matrix

$$\mathbf{H} = \hat{\mathbf{W}}^{\frac{1}{2}} \mathbf{X} \left(\mathbf{X}' \hat{\mathbf{W}} \mathbf{X}\right)^{-1} \mathbf{X}' \hat{\mathbf{W}}^{\frac{1}{2}}, \text{ where}$$

$$\hat{\mathbf{W}} = \text{diag}\left(\widehat{\pi}_i \left(1 - \widehat{\pi}_i\right)\right)$$

$$\mathbf{X} = \left(1, X_{i1}, X_{i2}, \ldots, X_{i,p-1}\right).$$ The **Design Matrix** (设计矩阵)

诊断：残差分析

Deviance residuals, the signed square root of the contribution of Y_i to the model deviance

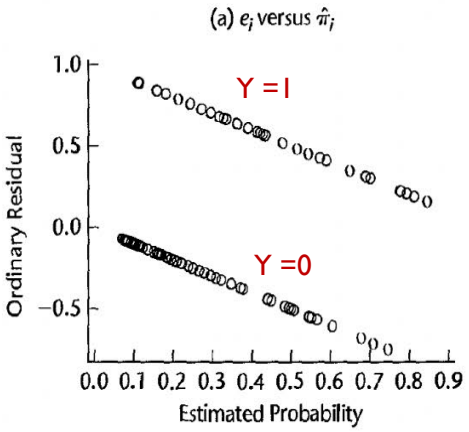
$$dev_i = \text{sign}\,(Y_i - \widehat{\pi}_i)\sqrt{-2\left[Y_i \log\frac{\widehat{\pi}_i}{Y_i} + (1-Y_i)\log\frac{1-\widehat{\pi}_i}{1-Y_i}\right]}$$
$$= \text{sign}\,(Y_i - \widehat{\pi}_i)\sqrt{-2\left[Y_i \log\widehat{\pi}_i + (1-Y_i)\log(1-\widehat{\pi}_i)\right]}$$

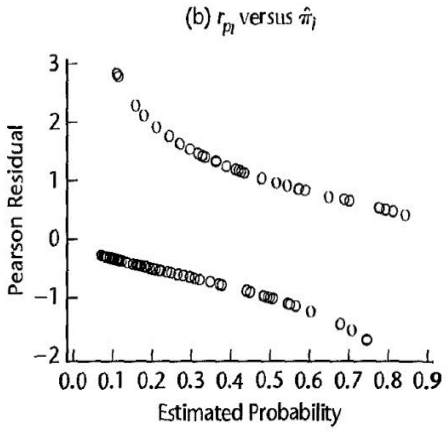
Analysis of residuals

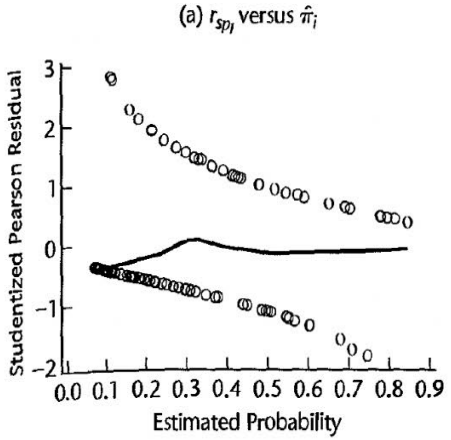
- to assess model inadequacy
- unknown distribution of residuals under true model
- plot residual by predicted value. A flat lowess smooth to this plot suggests good model
- better to group the data first

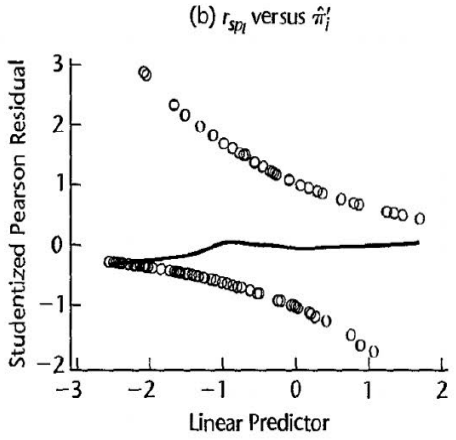
诊断：残差分析

Response Residual	$y_i - n_i\hat{\pi}_i$
Pearson Residual	$\frac{y_i - n_i\hat{\pi}_i}{\sqrt{n_i\hat{\pi}_i(1-\hat{\pi}_i)}}$
Deviance Residual	$\text{sign}(y_i - n_i\hat{\pi}_i)\sqrt{DEV_i}$
Studentized Pearson Residual	$\frac{y_i - n_i\hat{\pi}_i}{\sqrt{n_i\hat{\pi}_i(1-\hat{\pi}_i)\cdot(1-h_{ii})}}$

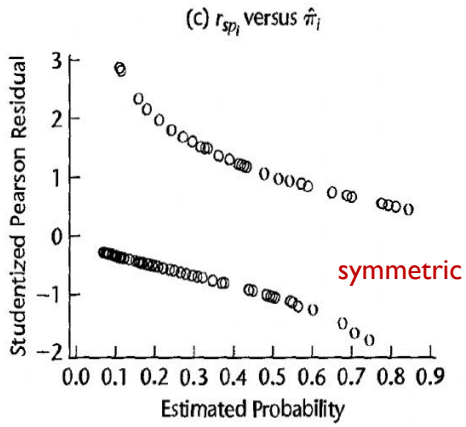


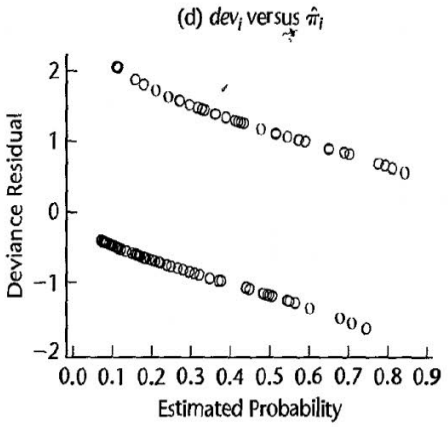


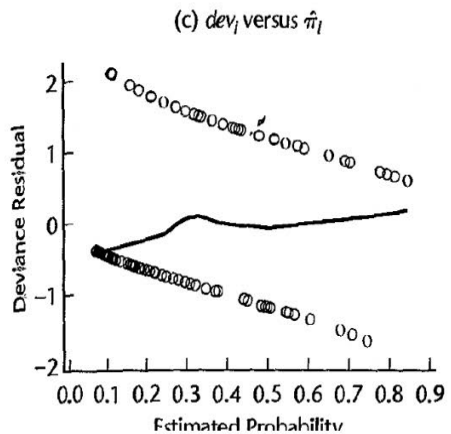


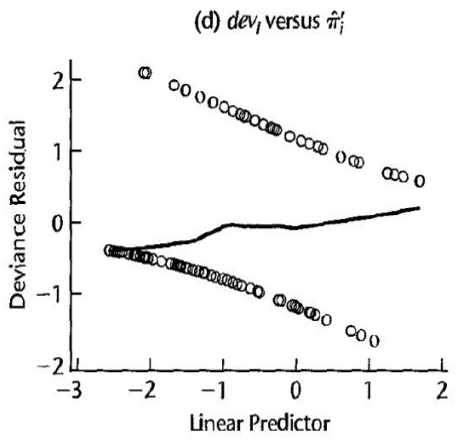


局部平滑拟合线（LOWESS 实线）









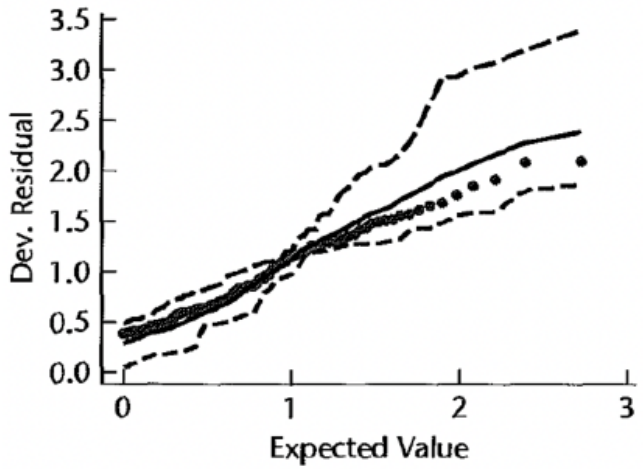
拟合良好的标准：平滑曲线应该大致水平，且紧贴 0 值线波动。

拟合不足的预警：如果平滑曲线出现明显的趋势、翘起或弯曲，则强烈暗示模型存在非线性关系未被捕捉

诊断：残差分析

Half-normal probability plot: plot ordered kth absolute standardized/deviance residuals against normal percentile,

$$z\left(\frac{k+n-1/8}{2n+1/2}\right)$$



Quantile-quantile plot (qqplot)

sample quantiles vs theoretical quantiles

Ordered Absolute residual vs absolte N(0,1)

To check for nonlinearity, one can plot the working residuals against covariates

诊断：残差分析


```r
# 1. 模拟标准二分类数据集
set.seed(42)
n <- 200
X_std <- rnorm(n, mean = 0, sd = 1)
# 真实的 Log-odds 是纯线性的
p_std <- 1 / (1 + exp(-(0.5 + 1.2 * X_std)))
Y_std <- rbinom(n, size = 1, prob = p_std)
data_std <- data.frame(X = X_std, Y = Y_std)

# 2. 拟合标准逻辑回归模型
model_std <- glm(Y ~ X, data = data_std, family = binomial(link = "logit"))

# 3. 提取四种残差
# 响应残差
res_response <- residuals(model_std, type = "response")
# Pearson 残差
res_pearson <- residuals(model_std, type = "pearson")
# 学生化 Pearson 残差 (利用 hatvalues)
h_ii <- hatvalues(model_std)
res_studentized <- res_pearson / sqrt(1 - h_ii)
# 偏差残差
res_deviance <- residuals(model_std, type = "deviance")

residuals_df <- data.frame(
  Response = head(res_response),
  Pearson = head(res_pearson),
  Studentized = head(res_studentized),
  Deviance = head(res_deviance)
)
print(residuals_df)
```

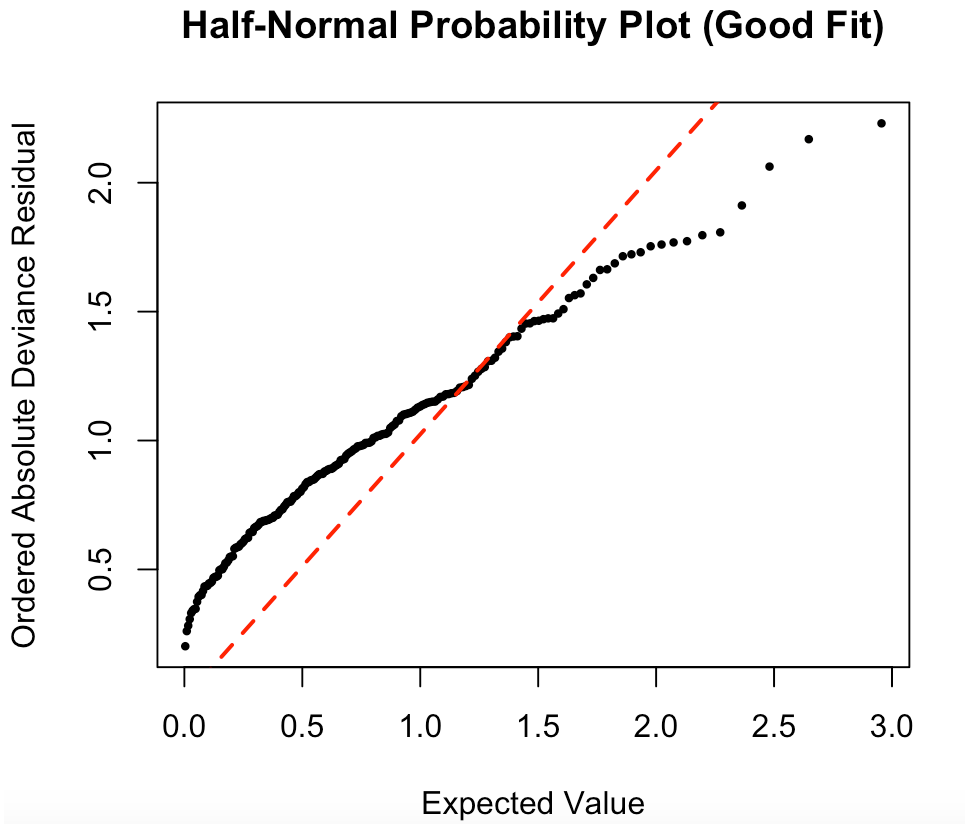
```
> residuals_df <- data.frame(
+   Response = head(res_response),
+   Pearson = head(res_pearson),
+   Studentized = head(res_studentized),
+   Deviance = head(res_deviance)
+ )
> print(residuals_df)
     Response    Pearson Studentized   Deviance
1  0.09750791  0.3286989   0.3306471  0.4529798
2 -0.45428714 -0.9123955  -0.9162766 -1.1006020
3  0.27465010  0.6153414   0.6177530  0.8013752
4  0.21302385  0.5202754   0.5227041  0.6921811
5 -0.73542966 -1.6672472  -1.6739629 -1.6307349
6  0.40437897  0.8239660   0.8266349  1.0179889
```

诊断：残差分析


```r
# 4. 绘制半正态概率图 (Half-normal probability plot)
abs_dev <- sort(abs(res_deviance))
k <- 1:n
# 按照课件公式计算期望的 Z 分位数
expected_quantiles <- qnorm((k + n - 1/8) / (2 * n + 1/2))

plot(expected_quantiles, abs_dev,
     xlab = "Expected Value", ylab = "Ordered Absolute Deviance Residual",
     main = "Half-Normal Probability Plot (Good Fit)",
     pch = 16, cex = 0.6)

# 添加参考直线
fit_line <- lm(abs_dev ~ expected_quantiles - 1)
abline(fit_line, col = "red", lty = 2, lwd = 2)
```

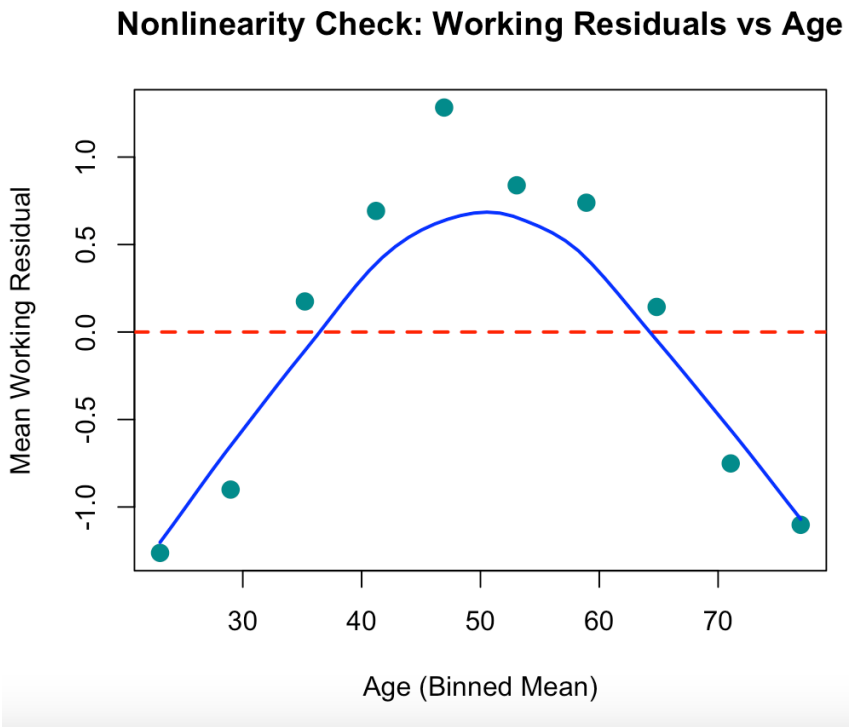


诊断：残差分析


```r
# ==========================================
# 1. 模拟非线性关系数据集（年龄与疾病风险）
# ==========================================
set.seed(888)
n <- 1000
age <- runif(n, 20, 80)

# 设定真实的 Log-odds 为一个开口向下的抛物线，顶点在 age = 50
# 数学公式：log_odds = -10 + 0.4 * age - 0.004 * age^2
true_log_odds <- -10 + 0.4 * age - 0.004 * age^2
p <- 1 / (1 + exp(-true_log_odds))
disease <- rbinom(n, size = 1, prob = p)

data <- data.frame(age = age, disease = disease)
```




```r
# ==========================================
# 2. 拟合模型（故意错误设定为纯线性）
# ==========================================
# 我们在模型中只放入 age 的一次项
model_linear <- glm(disease ~ age, data = data, family = binomial(link = "logit"))

# ==========================================
# 3. 提取工作残差并绘制诊断图
# ==========================================
# 提取工作残差 (Working Residuals)
res_working <- residuals(model_linear, type = "working")

# 核心教学步骤：数据分组 (Binning)
# 为了克服二分类点在图中上下聚集导致 LOWESS 曲线失真的问题，我们将年龄分为 10 个区间
age_bins <- cut(data$age, breaks = 10)
# 计算每个区间内 年龄的均值 和 工作残差的均值
bin_centers <- tapply(data$age, age_bins, mean)
bin_res_mean <- tapply(res_working, age_bins, mean)

# 绘制分组工作残差与协变量的散点图
plot(bin_centers, bin_res_mean,
     pch = 16, cex = 1.5, col = "darkcyan",
     xlab = "Age (Binned Mean)",
     ylab = "Mean Working Residual",
     main = "Nonlinearity Check: Working Residuals vs Age")

# 添加 0 水平参考线（完美拟合时，点应随机分布在该线两侧）
abline(h = 0, col = "red", lty = 2, lwd = 2)

# 添加 LOWESS 局部平滑曲线以捕捉全局趋势
lines(loess.smooth(bin_centers, bin_res_mean, span = 0.8),
      col = "blue", lwd = 2)
```

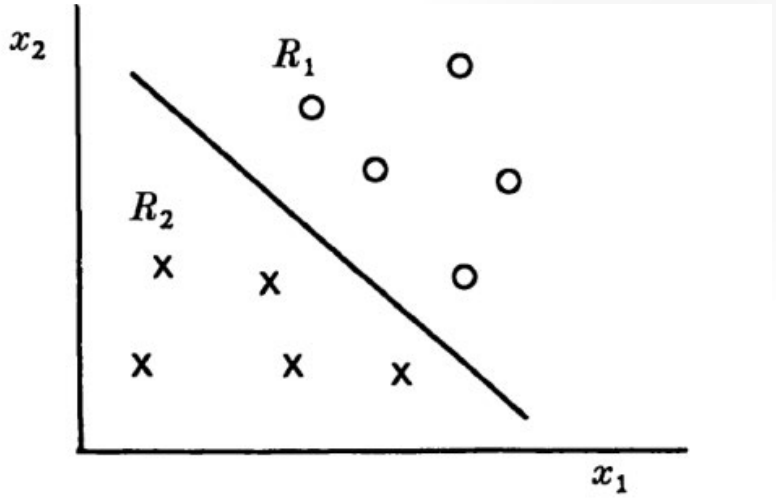
在回归公式中引入二次项age^2

完全分离 (Complete Separation)

- Complete separation: $b^T x_i > 0$ when $y_i = 1$, $b^T x_i < 0$ when $y_i = 0$.

  centering by (β_0)
- For $\beta = kb$, as $k \to \infty$, the value of likelihood function increases to 1.

  MLE doesn't exist
- Indications:
  - huge standard errors of $\hat{\beta}$
  - fitted values close to y (0 or 1)
  - maximum log-likelihood close to 0



Derivation

Under logistic regression, $\pi_i(\beta) = P(Y_i = 1 \mid x_i) = \frac{e^{x_i^\top \beta}}{1+e^{x_i^\top \beta}}$, and the log-likelihood is

$$\ell(\beta) = \sum_{i=1}^{n} \Big[y_i \log \pi_i(\beta) + (1 - y_i) \log\big(1 - \pi_i(\beta)\big) \Big].$$

For any finite β, $0 < \pi_i(\beta) < 1$, so $\log \pi_i(\beta) < 0$, $\log(1 - \pi_i(\beta)) < 0$, and therefore $\ell(\beta) < 0$.

Take $\beta = kb$ with $k \to \infty$. Then $x_i^\top \beta = k\, b^\top x_i$. Hence

$$x_i^\top \beta \to +\infty \quad \text{if } y_i = 1, \qquad x_i^\top \beta \to -\infty \quad \text{if } y_i = 0.$$

Therefore,

$$\pi_i(\beta) \to 1 \quad \text{if } y_i = 1, \qquad \pi_i(\beta) \to 0 \quad \text{if } y_i = 0.$$

So each term in the log-likelihood tends to 0:

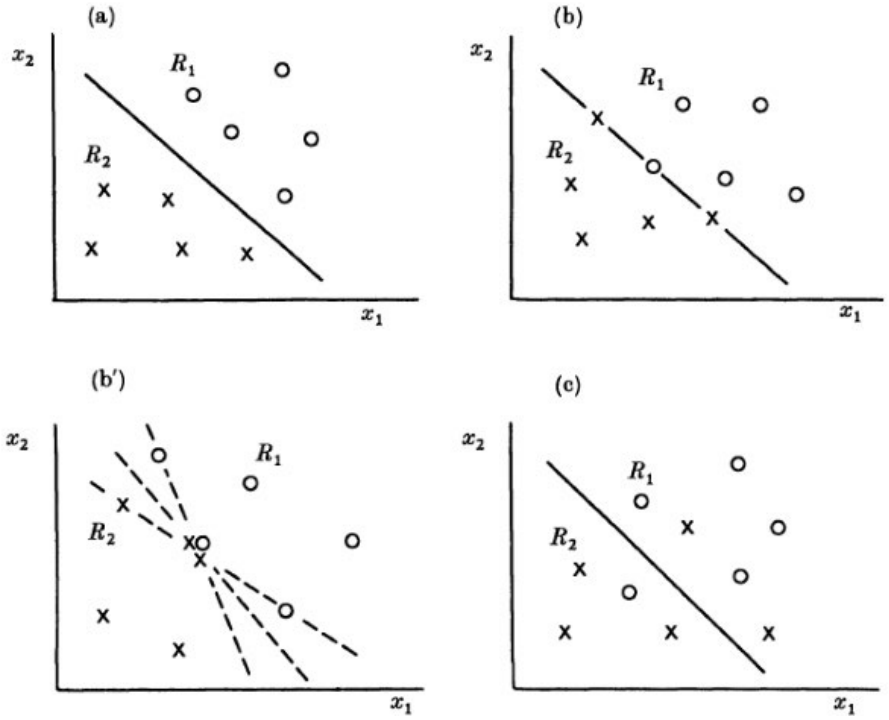
$$y_i \log \pi_i(\beta) + (1 - y_i) \log\big(1 - \pi_i(\beta)\big) \to 0.$$

Thus $\ell(kb) \to 0$.

准完全分离 (Quasi-complete separation)

- Quasi-complete separation: $b^T x_i \geq 0$ when $y_i = 1$, $b^T x_i \leq 0$ when $y_i = 0$, both inequality holds for at least one case.

- Indications:
  - huge standard errors of $\hat{\beta}$
  - fitted values close to y (0 or 1) for some observations
  - maximum log-likelihood less than 0



过度离散 (Over-Dispersion)

A binomial r.v. is a sum of *i.i.d.* Bernoulli r.v.'s. For m large, the components may not be *i.i.d.*, and over-dispersion may occur.

Suppose $Y = \Sigma_{i=1}^{c} X_i$, $X_i \sim Binomial(n_i, pi)$'s are independent. $\Sigma_{i=1}^{c} n_i = m, E[p_i] = p$, and $\text{var}(p_i) = \tau^2 p(1-p)$. It follows that

$$E[Y] = mp, \text{var}[Y] = mp(1-p)\left\{1 + \tau^2 \sum_{i=1}^{c} (n_i^2 - n_i)/m\right\}$$

Clustering is seen to be one possible cause of over-dispersion. A simple "model" for over-dispersion is to assume $\text{var}[Y] = v\sigma^2$ and estimate v by $\tilde{v} = X^2/(m-p)$

Multi-category Response: Nominal vs. Ordinal

Binary logistic: $Y \in \{0,1\}$. **What if Y has $J > 2$ categories?**

Two fundamentally different cases:

Type	Definition	Example	Appropriate Model
Nominal 名义型	No natural ordering	Blood type A/B/AB/O, brand choice, major field	Baseline-category logit
Ordinal 有序型	Categories have a meaningful order	Pain: None < Mild < Moderate < Severe	Cumulative logit (proportional odds)

Why does the distinction matter?

If ordinal Y is treated as **nominal**:

→ Ignores ordering information → **wastes statistical power**; $J-1$ times more parameters

If nominal Y is assumed to be **ordinal**:

→ Imposes false structure → **biased estimates and misleading predictions**

Nominal Response: Baseline-Category Logit Model

Choose category J as **baseline (reference)**. Model the log-odds of each category vs. baseline:

$$\log\left[\frac{P(Y=j)}{P(Y=J)}\right] = \alpha_j + \boldsymbol{\beta}_j^T \mathbf{x}, \quad j = 1, \ldots, J-1$$

Gives $J-1$ **simultaneous logistic regression equations** — each with its own $(\alpha_j, \boldsymbol{\beta}_j)$
Predicted probabilities via softmax:

$$P(Y=j|\mathbf{x}) = \frac{\exp(\alpha_j + \boldsymbol{\beta}_j^T \mathbf{x})}{1 + \sum_{k=1}^{J-1} \exp(\alpha_k + \boldsymbol{\beta}_k^T \mathbf{x})}$$

Example: Student major choice (J = 3: Arts / Science / Engineering as baseline)

$$\log[P(Arts)/P(Eng)] = \alpha_1 + \boldsymbol{\beta}_1{}^{\mathrm{T}}\mathbf{x} \quad \text{and} \quad \log[P(Sci)/P(Eng)] = \alpha_2 + \boldsymbol{\beta}_2{}^{\mathrm{T}}\mathbf{x}$$

Baseline choice does not affect model fit or predictions — only the interpretation of coefficients

R Example: Baseline-Category Logit (nnet)

Dataset: Graduate career choice

Y = career: Academia / Industry / Gov (baseline)

X: GPA, scholarship (0/1)

Key output:

exp(coef) = Odds Ratio vs. baseline

e.g. OR_Academia[GPA] = 2.4

→ one-unit ↑ GPA doubles odds of choosing Academia over Gov

predict(fit, type="probs")

returns $P(Y = j \mid x)$ for all J categories simultaneously

reflevel= controls the baseline; re-run with different baseline to get cross-comparisons (Academia vs. Industry)

```r
library(nnet)

# 模拟数据
set.seed(42)
n <- 300
grad <- data.frame(
  gpa         = round(runif(n, 2.0, 4.0), 2),
  scholarship = rbinom(n, 1, 0.4),
  career      = sample(c("Academia", "Industry", "Government"),
                       n, replace = TRUE, prob = c(0.3, 0.5, 0.2))
)

grad$career <- relevel(factor(grad$career), ref = "Government")

# 拟合模型
fit <- multinom(career ~ gpa + scholarship, data = grad)

# 查看结果
summary(fit)
exp(coef(fit))                    # Odds Ratios vs. baseline
predict(fit, type = "probs") |> head()   # 预测概率
```

R Example: Baseline-Category Logit

Dataset: Graduate career choice

Y = career: Academia / Industry / Gov (baseline)

X: GPA, scholarship (0/1)

Key output:

exp(coef) = Odds Ratio vs. baseline

gpa OR = 0.513 (Academia), 0.609 (Industry)

→ both < 1, consistent with above

predict(fit, type="probs")

returns $P(Y = j \mid x)$ for all J categories simultaneously

reflevel= controls the baseline; re-run with different baseline to get cross-comparisons (Academia vs. Industry)

```
> summary(fit)
Call:
multinom(formula = career ~ gpa + scholarship, data = grad)

Coefficients:
          (Intercept)        gpa scholarship
Academia    2.614394 -0.6681613  -0.5550903
Industry    2.601086 -0.4955692  -0.3283949

Std. Errors:
          (Intercept)       gpa scholarship
Academia   0.9458274 0.2978833   0.3554397
Industry   0.8677299 0.2695525   0.3157573

Residual Deviance: 605.3333
AIC: 617.3333
> exp(coef(fit))                    # Odds Ratios vs. baseline
          (Intercept)       gpa scholarship
Academia    13.65893 0.5126503   0.5740204
Industry    13.47836 0.6092240   0.7200786
> predict(fit, type = "probs") |> head()   # 预测概率
  Government  Academia  Industry
1  0.2452898 0.2592456 0.4954646
2  0.2494206 0.2566594 0.4939200
3  0.1384223 0.3395183 0.5220595
4  0.2282329 0.2702344 0.5015327
5  0.2641074 0.2313866 0.5045060
6  0.2392356 0.2460517 0.5147128
```

Ordinal Response: Why Ordering Matters

Running example: post-surgery pain assessment

Y : *None (1)* < *Mild (2)* < *Moderate (3)* < *Severe (4)*

Predictors: age, treatment group, baseline pain score

Why not treat as nominal (baseline-category logit)?

→ Ignores ordering entirely — "None" and "Severe" treated as equally different from "Mild"

→ **Wastes information**; requires $3 \times p$ parameters for $J = 4$ instead of just p

Why not treat as continuous (linear regression)?

→ Assumes equal spacing: gap (None → Mild) = gap (Mild → Moderate) = gap (Moderate → Severe)

→ **Predicted values may fall outside {1, 2, 3, 4}**, e.g. $\hat{y} = 1.7$ or $\hat{y} = 4.3$

Ordinal models use the ordering WITHOUT assuming equal spacing

Cumulative Logit Model (Proportional Odds)

Model **cumulative probabilities** $P(Y \leq j)$ for each of $J-1$ cut points:

$$\log\left[\frac{P(Y\leq j)}{P(Y>j)}\right] = \alpha_j - \boldsymbol{\beta}^\top \mathbf{x}, \quad j = 1, \ldots, J-1$$

Key: β is **SHARED** across all $J-1$ equations — **Proportional Odds assumption**

Intercepts only: $\alpha_1 < \alpha_2 < \cdots < \alpha_{j-1}$

Pain example (J = 4) — two equations, SAME β_1 and β_2:

$$\log[\, P(pain \leq mild) \,/\, P(pain > mild) \,] = \alpha_1 - \beta_1 \cdot age - \beta_2 \cdot treatment$$

$$\log[\, P(pain \leq moderate) \,/\, P(pain > moderate) \,] = \alpha_2 - \beta_1 \cdot age - \beta_2 \cdot treatment$$

$\rightarrow$ Same β! One $\exp(\beta_j)$ per predictor = **proportional odds ratio**

Interpreting the Cumulative Logit Model

Intercepts: $\alpha_1 < \alpha_2 < \ldots < \alpha_{j-1}$ — always increasing cut points on the latent scale

Coefficient β: one β per predictor, shared across all cut points

$\beta > 0 \rightarrow$ higher $x \rightarrow$ higher $P(Y > j) \rightarrow$ shifted toward larger (more severe) categories

$\exp(\beta)$ = proportional odds ratio (与logistic的 OR 解释相同)

Model comparison summary :

Model	Response Y	No. of Equations	Shared β?
Binary logistic	Binary (J = 2)	1 logit	N/A
Baseline-category logit	Nominal (J > 2)	J−1 (one per category)	No — β_j per category
Cumulative logit	Ordinal (J > 2)	J−1 (one per cut point)	Yes — proportional odds

R Example: Cumulative Logit

Dataset: Post-surgery pain assessment

Y = pain (ordered factor):

None < Mild < Moderate < Severe

```r
library(MASS)
library(ordinal)

# 模拟数据
set.seed(42)
n <- 200
pain <- data.frame(
  age       = round(runif(n, 20, 80)),
  treatment = rbinom(n, 1, 0.5)
)

latent <- -0.03 * pain$age + 0.8 * pain$treatment + rnorm(n)
pain$pain <- factor(
  cut(latent, breaks = quantile(latent, probs = c(0, 0.25, 0.5, 0.75, 1)),
      labels = c("None", "Mild", "Moderate", "Severe"),
      include.lowest = TRUE),
  ordered = TRUE)

# 拟合
fit <- polr(pain ~ age + treatment, data = pain, Hess = TRUE)
summary(fit)

# 比例优势比 + p 值
ctab <- coef(summary(fit))
cbind(OR      = round(exp(ctab[, "Value"]), 3),
      p_value = round(pnorm(abs(ctab[, "t value"]),
                            lower.tail = FALSE) * 2, 4))

# 检验比例优势假设
nominal_test(clm(pain ~ age + treatment, data = pain))
```

R Example: Cumulative Logit

Dataset: Post-surgery pain assessment

Y = pain (ordered factor):

None < Mild < Moderate < Severe

Key output:

Intercepts = $\alpha_1 < \alpha_2 < \alpha_3$

(cut points on latent scale)

age: OR = 0.948, p < 0.001 → each additional year of age reduces odds of being in a higher pain category by 5.2%

treatment: OR = 3.257, p < 0.001 → treatment group has 3× higher odds of worse pain (simulated direction)

brant(fit) tests the proportional odds assumption:

p > 0.05 → assumption holds

```
> summary(fit)
Call:
polr(formula = pain ~ age + treatment, data = pain, Hess = TRUE)

Coefficients:
             Value Std. Error t value
age       -0.05369   0.008338  -6.439
treatment  1.18069   0.273525   4.317

Intercepts:
                Value   Std. Error t value
None|Mild       -3.6108  0.4903    -7.3638
Mild|Moderate   -2.2857  0.4533    -5.0420
Moderate|Severe -0.9006  0.4223    -2.1325

Residual Deviance: 495.6783
AIC: 505.6783
>
> # 比例优势比 + p 值
> ctab <- coef(summary(fit))
> cbind(OR      = round(exp(ctab[, "Value"]), 3),
+       p_value = round(pnorm(abs(ctab[, "t value"]),
+                             lower.tail = FALSE) * 2, 4))
                  OR p_value
age            0.948   0.000
treatment      3.257   0.000
None|Mild      0.027   0.000
Mild|Moderate  0.102   0.000
Moderate|Severe 0.406  0.033
>
> # 检验比例优势假设
> nominal_test(clm(pain ~ age + treatment, data = pain))
Tests of nominal effects

formula: pain ~ age + treatment
          Df  logLik    AIC    LRT Pr(>Chi)
<none>       -247.84 505.68
age        2 -247.28 508.56 1.1184   0.5717
treatment  2 -246.59 507.17 2.5044   0.2859
```

Poisson Regression: Modeling Count Data

When $Y = count\ (\#\ of\ events),\ Y \in \{0, 1, 2, \dots\}$:
$Y \sim Poisson(\lambda)$, $V(\mu) = \mu$

$$\log(\lambda_i) = \mathbf{x}_i^\top \boldsymbol{\beta} \implies \lambda_i = \exp(\mathbf{x}_i^\top \boldsymbol{\beta}) > 0$$

Coefficient interpretation:

$\exp(\beta_j)$ = **multiplicative effect** on expected count

e.g. $\exp(\beta_1) = 1.3 \rightarrow$ one-unit increase in $x_1 \rightarrow$ **30% increase in λ**

Applications Example:

Hospital readmissions: # of readmissions in 30 days ~ age, diagnosis, treatment

Traffic accidents: # of accidents/month at intersection ~ volume, speed limit

Log-Linear Models: Associations in Contingency Tables

Apply Poisson regression to **cell counts** in a contingency table.
Goal: model the **association structure** between categorical variables.
All variables are treated symmetrically — no single designated response

Example: Smoking × Lung Disease (2 × 2 contingency table)

$$\log(\mu_{ij}) = \mu + \lambda_i^S + \lambda_j^D + \lambda_{ij}^{SD}$$

μ : overall log-mean
λ_i^S : main effect of Smoking
λ_j^D : main effect of Lung Disease
$\lambda_{ij}^{SD} = 0$ for all i, j $\Leftrightarrow$ **conditional independence** between Smoking and Disease

Connection to logistic regression: When one variable is a binary response → log-linear model is **equivalent to logistic regression** (same likelihood, same estimates)

Common applications: epidemiology, market research, survey data

R Example: Poisson Regression

Dataset: Monthly traffic accidents

Y = accidents (count, ≥ 0)

X: volume, speed_limit, lighting

```r
set.seed(42)
n <- 300
traffic <- data.frame(
  traffic_volume = rpois(n, lambda = 1000),
  speed_limit    = sample(c(40, 60, 80), n, replace = TRUE),
  lighting       = rbinom(n, 1, 0.6)   # 1 = 有路灯
)

# 生成事故次数 (log-linear 结构)
log_lambda <- 0.5 +
  0.001  * traffic$traffic_volume +
  0.015  * traffic$speed_limit -
  0.5    * traffic$lighting
traffic$accidents <- rpois(n, lambda = exp(log_lambda))

# 拟合 Poisson 回归
fit <- glm(accidents ~ traffic_volume + speed_limit + lighting,
           data   = traffic,
           family = poisson(link = "log"))

summary(fit)
exp(coef(fit))        # Rate Ratios
exp(confint(fit))     # 95% CI

# 过度离散诊断 (> 1 说明存在过度离散)
fit$deviance / fit$df.residual

# 若过度离散，改用负二项回归
fit_nb <- glm.nb(accidents ~ traffic_volume + speed_limit + lighting,
                 data = traffic)
summary(fit_nb)
```

R Example: Poisson Regression


```
> summary(fit)

Call:
glm(formula = accidents ~ traffic_volume + speed_limit + lighting,
    family = poisson(link = "log"), data = traffic)

Coefficients:
                Estimate Std. Error z value Pr(>|z|)
(Intercept)     1.0531150  0.6753227   1.559    0.119
traffic_volume  0.0003757  0.0006672   0.563    0.573
speed_limit     0.0167060  0.0012397  13.476   <2e-16 ***
lighting       -0.5550661  0.0391390 -14.182   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for poisson family taken to be 1)

    Null deviance: 738.96  on 299  degrees of freedom
Residual deviance: 330.71  on 296  degrees of freedom
AIC: 1499.2

Number of Fisher Scoring iterations: 4

> exp(coef(fit))        # Rate Ratios
   (Intercept) traffic_volume    speed_limit       lighting
     2.8665667      1.0003757      1.0168463      0.5740343
> exp(confint(fit))     # 95% CI
Waiting for profiling to be done...
                   2.5 %     97.5 %
(Intercept)    0.7627592 10.7661283
traffic_volume 0.9990679  1.0016841
speed_limit    1.0143853  1.0193273
lighting       0.5316134  0.6197757
```

Key output:

exp(coef) = Rate Ratio（倍率比）

exp(β_lighting) = 0.57
→ good lighting → 43% fewer accidents expected

Overdispersion check: if deviance/df ≫ 1,
switch to **glm(..., family=quasipoisson)**
or **MASS::glm.nb()** (Negative Binomial)

R Example: Log-Linear Model

Dataset: UCBAdmissions (built-in R)

3 variables: Admit × Gender × Dept

Y = cell count Freq

Two models:

fit0: no interaction (independence)

fit1: + Gender × Dept interaction

anova() LRT:

$p < 0.05 \rightarrow$ Gender & Dept NOT independent

exp(coef) interaction term:

= association ratio between variables

Saturated model (all interactions) fits table exactly — use AIC/LRT for simpler models

```r
data(UCBAdmissions)
df <- as.data.frame(UCBAdmissions)
head(df)

# 独立性模型（无交互项）
fit0 <- glm(Freq ~ Admit + Gender + Dept,
            data   = df,
            family = poisson)

# 含 Gender × Dept 关联项的模型
fit1 <- glm(Freq ~ Admit + Gender + Dept + Gender:Dept,
            data   = df,
            family = poisson)

# 似然比检验（LRT）
anova(fit0, fit1, test = "Chisq")

# 关联强度（交互项的 exp(coef)）
exp(coef(fit1))

# AIC 比较
AIC(fit0, fit1)
```

```
> head(df)
      Admit Gender Dept Freq
1 Admitted   Male    A  512
2 Rejected   Male    A  313
3 Admitted Female    A   89
4 Rejected Female    A   19
5 Admitted   Male    B  353
6 Rejected   Male    B  207
```

R Example: Log-Linear Model

Dataset: UCBAdmissions (built-in R)

3 variables: Admit × Gender × Dept

Y = cell count Freq

Two models:

fit0: no interaction (independence)

fit1: + Gender × Dept interaction

anova() LRT:

$p < 0.05 \rightarrow$ Gender & Dept NOT independent

exp(coef) interaction term:

= association ratio between variables

Saturated model (all interactions) fits table exactly — use AIC/LRT for simpler models

```
> # 似然比检验（LRT）
> anova(fit0, fit1, test = "Chisq")
Analysis of Deviance Table

Model 1: Freq ~ Admit + Gender + Dept
Model 2: Freq ~ Admit + Gender + Dept + Gender:Dept
  Resid. Df Resid. Dev Df Deviance  Pr(>Chi)
1        16    2097.67
2        11     877.06  5   1220.6 < 2.2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
>
> # 关联强度（交互项的 exp(coef)）
> exp(coef(fit1))
       (Intercept)      AdmitRejected       GenderFemale              DeptB
       319.9016792          1.5789174          0.1309091          0.6787879
             DeptC              DeptD              DeptE              DeptF
         0.3939394          0.5054545          0.2315152          0.4521212
GenderFemale:DeptB GenderFemale:DeptC GenderFemale:DeptD GenderFemale:DeptE
         0.3410218         13.9380342          6.8695044         15.7177138
GenderFemale:DeptF
         6.9835419
>
> # AIC 比较
> AIC(fit0, fit1)
     df      AIC
fit0  8 2272.731
fit1 13 1062.116
```

GLM Family: One Framework, Many Models

GLM is not a single model — it is a **language** for building models tailored to the nature of Y.

Model	Distribution	Link g(μ)	Use Case
Linear regression	Normal	Identity: g(μ) = μ	Continuous Y, unbounded
Logistic regression	Bernoulli	Logit: log(μ/(1−μ))	Binary Y ∈ {0,1}
Multinomial logit	Multinomial	Baseline-category logit	Nominal Y, J > 2
Cumulative logit	Multinomial	Cumulative logit	Ordinal Y, J > 2
Poisson regression	Poisson	Log: log(μ)	Count Y ∈ {0,1,2,...}
Log-linear model	Poisson	Log: log(μ)	Contingency table counts
Gamma regression	Gamma	Log or Reciprocal	Positive, right-skewed Y

Decision rule: ① What type is Y ? → choose distribution family → ② Canonical link or custom? → specify $g(\mu)$ → model fully defined